Khalil Peguero

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Education

Florida International University — Miami, FL
Bachelor of Science in Computer Science — *Expected Summer 2027*
- **Relevant Coursework:** Data Structures & Algorithms, Algorithm Techniques, Systems Programming, Programming Languages & Compiler Theory, Data Mining & Machine Learning, Probability & Statistics

Technical Skills

Languages: Python, Go, Java, C#, C, SQL, TypeScript, JavaScript, HTML/CSS
Frameworks & Tools: FastAPI, ASP.NET/.NET, React, Node.js, pytest, Git, n8n
Data & Infrastructure: PostgreSQL, MySQL, SQLite, Redis, Docker, Linux, CI/CD, Solana
Engineering Practices: REST API design, microservices, automated testing (unit, integration, mutation), JWT/OAuth2

Experience

Sigma Lambda Beta Education Foundation, Inc. — Remote
Software Engineer — *May 2025 – Present*
- Maintain an **ASP.NET/C#** and **MySQL** donation portal (**27K+** donations) with **Authorize.NET** payments.
- Consolidated payment types via an idempotent, self-validating **SQL** migration with rollback, tested against **24** scenarios and a production snapshot; fixed a bug where editing a contribution silently overwrote its payment type.
- Refactored the District model into Region across the **MySQL** schema, entity layer, services and reports (**30+** files), fixing **2** pre-existing crashes found in QA and halving the region report's database round-trips.
- Authored **C#** unit tests for the contribution validation service (**98.7%** line coverage) and integration tests for bug fixes, run by **CI** on every commit; peer-reviewed **12** teammate PRs spanning **PCI** remediation and **OAuth** flows.

BNY — Remote
Sophomore Summit – Tech Lead — *May 2025 – July 2025*
- Led a **5-member team** in a mock-IPO coding challenge; designed and presented a **Python** share-allocation algorithm based on bid price, timestamp, and demand to senior engineers, refining the solution from their feedback.

Domus Care Solutions — Weston, FL
Software Engineer Intern — *May 2024 – July 2024*
- Built a **Java Swing** desktop report builder that formats sensitive data for **hundreds of patients** into internal reports.
- Automated **Google Sheets** dashboards with data analysts to track patient activity, billing time, and insurance claims, cutting manual data-sorting by **8+ hours per week**.

Projects

MomentumX — Tokenized Stock Trading Platform | *Python, FastAPI, SQLite, Solana, pytest* — September 2026
- Won **MLH Best Use of Solana** at **ShellHacks 2026** (294 projects) as backend lead of a 4-person team; designed the **REST API** contracts and **SQLite** schema.
- Built the **Solana** transaction layer in **FastAPI**: verifies each wallet-signed trade against its quote, then co-signs and confirms it; made submission **idempotent** so dropped connections never lose or double a trade.
- Designed an average-cost P/L ledger that serves portfolio stats in ∼**11 ms**; fixed intermittent **HTTP 500s** under polling via per-thread SQLite connections and **WAL**, cutting load-test errors from **50/900 to 0**.

QuantSim — Paper Trading & Backtesting Platform | *Go, PostgreSQL, Redis, Docker, React* — February 2026 – Present
- Architected and built a **Go** microservices platform (auth, API gateway, market data, trading, backtesting, analytics) on **PostgreSQL** and **Redis**, shipped as distroless **Docker** images; **600+ tests** include integration suites on real Postgres.
- Prevented double-spends on concurrent orders with a row-locked (**SELECT FOR UPDATE**) Postgres transaction; hardened auth with JWT refresh-token revocation, per-IP rate limiting and exponential login backoff.
- Built MA/RSI/MACD backtests with look-ahead-safe fills and Sharpe/drawdown metrics.
- Added **Claude API** portfolio summaries with a hallucination guard: the model writes prose only, Go injects every figure, and drafts containing model-generated numbers are rejected.

HermeΣ — Chapter Automation Platform | *n8n, Python, Slack/Google Calendar APIs, Linux VPS* — May 2026 – Present
- Build and maintain a **Slack**-integrated platform that turns weekly chapter minutes into summaries, task lists and **Google Calendar** events for **20+ members** across **2 chapters**, hosted on a **Linux VPS**.
- Built a **diff-based sync** that reconciles events extracted from meeting minutes against **Google Calendar** (create, update, delete) with recurring-event deduplication; added a natural-language **Slack** calendar agent restricted to E-Board members.

Leadership & Involvement

- **VP of Finance** (2024–Present) and **President** (2024–2025), FIU chapter of Sigma Lambda Beta International Fraternity; coordinated a joint emergency fundraiser with Sigma Lambda Gamma for Venezuela earthquake relief.
- **Member**, INIT and SHPE